Homework Set 6

Problem 1: Recall that in the policy gradient algorithms like REINFORCE, actor-critic, and advantage actor-critic where we search over a parameterized class of policies

$$\Pi = \{\pi_\theta : \theta \in \mathbb{R}^d\},$$

the gradient of the objective function (value function) relies on computing $\nabla_\theta \log \pi_\theta(a_t|s_t)$ for sampled state-action pair (s_t, a_t). Notice that $\pi_\theta(a|s)$ is a function over state-action pairs, i.e., $\pi_\theta : \mathcal{S} \times \mathcal{A} \to (0, 1)$ that is parameterized by $\theta \in \mathbb{R}^d$.

1. Compute $\nabla_\theta \log \pi_\theta(a_t|s_t)$ for the case of using the class of softmax policies

$$\forall s \in \mathcal{S}, \forall a \in \mathcal{A} : \quad \pi_\theta(a|s) = \frac{\exp(\theta_{s,a})}{\sum_{a' \in \mathcal{A}} \exp(\theta_{s,a'})},$$

   where $\theta \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$.
   [Hint: You need to compute each entry $\frac{\partial \log \pi_\theta(a_t|s_t)}{\partial \theta_{s,a}}$ of the gradient vector $\nabla_\theta \log \pi_\theta(a_t|s_t) \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$. It would be helpful to partition these entries into three groups:
   1) $s \neq s_t$, 2) $s = s_t, a \neq a_t$, and 3) $s = s_t, a = a_t$.]

2. Compute $\nabla_\theta \log \pi_\theta(a_t|s_t)$ for the case of using the class of softmax linear policies

$$\forall s \in \mathcal{S}, \forall a \in \mathcal{A} : \quad \pi_\theta(a|s) = \frac{\exp(\theta^\top \phi(s,a))}{\sum_{a' \in \mathcal{A}} \exp(\theta^\top \phi(s,a'))},$$

   where $\theta \in \mathbb{R}^d$ and $\phi(s,a) \in \mathbb{R}^d$.
   [Hint: It may be helpful to write $\theta^\top \phi(s,a) = \sum_{l=1}^{d} \theta_l \phi_l(s,a)$, where θ_l and $\phi_l(s,a)$ denote the l^{th} entry of θ and $\phi(s,a)$, respectively.]

3. Compute $\nabla_\theta \log \pi_\theta(a_t|s_t)$ for the case of using the class of softmax neural policies

$$\forall s \in \mathcal{S}, \forall a \in \mathcal{A} : \quad \pi_\theta(a|s) = \frac{\exp(f_\theta(s,a))}{\sum_{a' \in \mathcal{A}} \exp(f_\theta(s,a'))},$$

   where $\theta \in \mathbb{R}^d$ and $f_\theta(s,a) : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$. Assume that $f_\theta(s,a)$ is a differentiable function and keep your solution in terms of its partial derivatives.

4. Consider searching for an optimal policy over a parameterized class of softmax policies

$$\Pi = \{\pi_\theta : \pi_\theta(a|s) = \frac{\exp\left(\theta_{s,a}\right)}{\sum_{a' \in \mathcal{A}} \exp\left(\theta_{s,a'}\right)} \; \textit{for all } s \in \mathcal{S}, a \in \mathcal{A} \textit{ and } \theta \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}\},$$

where $\mathcal{S} = \{1, 2, \ldots, 100\}$ and $\mathcal{A} = \{b, c, d, e\}$, by applying the REINFORCE algorithm in an episodic discounted setting. At iteration i of the algorithm, we sample trajectory τ_i and form a stochastic gradient

$$g_i = \left(\sum_{t=0}^{T} \nabla_\theta \log\left(\pi_\theta(a_t^i|s_t^i)\right)\right)\left(\sum_{t=0}^{T} \gamma^t R(s_t^i, a_t^i)\right),$$

where s_t^i and a_t^i denote the state and action visited at time t in trajectory τ_i.

(a) Let the set of states, the set of actions, and the set of state-action pairs visited in trajectory τ_i be $\mathcal{S}^i \subseteq \mathcal{S}$, $\mathcal{A}^i \subseteq \mathcal{A}$. and $\mathcal{SA}^i \subseteq \mathcal{S} \times \mathcal{A}$, respectively. Which subset of parameters $\theta_{s,a}$ may change in the stochastic gradient ascent step? [Explain your reasoning.]

(b) Suppose the return of this trajectory is positive. Also, this trajectory visits state $s = 7$ a number of times and in that, it always takes action $a = d$. What will happen (decrease, stay the same, or increase) to parameters $\theta_{7,b}$ and $\theta_{7,d}$ in the stochastic gradient ascent step? [Explain your reasoning.]

Problem 2: In maximum entropy reinforcement learning (and inverse reinforcement learning), the reward is augmented with the entropy of the policy, i.e., the goal is to find a stochastic policy that maximizes

$$J(\pi) = \mathbb{E}_{\substack{S_0 \sim \mu_0 \\ A_t \sim \pi(.|S_t) \\ S_{t+1} \sim P(.|S_t, A_t)}} \left[\sum_{t=0}^{\infty} \gamma^t \left(R(S_t, A_t) + \lambda \mathcal{H}(\pi(.|S_t))\right)\right],$$

where $\lambda \geq 0$. Soft policy iteration algorithm is a variant of policy iteration algorithm that finds such an optimal stochastic policy through iterative policy evaluation and policy improvement. The policy evaluation step computes the state-action value function $Q^{\pi_t}(s, a)$ of the current policy π_t for all $s \in \mathcal{S}, a \in \mathcal{A}$ and the policy improvement step computes the next policy π_{t+1} using

$$\pi_{t+1}(.|s) = \arg\max_{\pi \in \Pi} \quad \mathbb{E}_{A \sim \pi(.|s)}\left[Q^{\pi_t}(s, A)\right] + \lambda \mathcal{H}(\pi(.|s)) \qquad \forall s \in \mathcal{S}.$$

Suppose the action space $\mathcal{A}$ is discrete and finite. The policy improvement step can be written as a constrained optimization problem

$$\begin{aligned} \max_{\pi(.|s) \geq \mathbf{0}} \quad & \mathbb{E}_{A \sim \pi(.|s)}\left[Q^{\pi_t}(s, A)\right] + \lambda \mathcal{H}(\pi(.|s)) \\ \textit{subject to} \quad & \sum_{a \in \mathcal{A}} \pi(a|s) = 1, \end{aligned}$$

and solved individually for each $s \in \mathcal{S}$. This constrained optimization problem is equivalent to

$$\max_{\pi(.|s) \geq \mathbf{0}} \min_{\beta \in \mathbb{R}} \quad \mathcal{L}(\pi, \beta),$$

where $\mathcal{L}(\pi, \beta)$ is called the Lagrangian function and is defined as

$$\mathcal{L}(\pi, \beta) = \mathbb{E}_{A \sim \pi(.|s)}\left[Q^{\pi_t}(s, A)\right] + \lambda \mathcal{H}(\pi(.|s)) - \beta(\sum_{a \in \mathcal{A}} \pi(a|s) - 1).$$

1. Find the derivative of the Lagrangian function with respect to $\pi(a|s)$ for each $a \in \mathcal{A}$.
2. Find the derivative of the Lagrangian function with respect to β.
3. Find the stationary point of the Lagrangian function, i.e., $\pi(a|s)$ and β where the derivatives (computed in Part 1 and Part 2) are zero. The answer should be based on the known values.
4. The resulting policy in Part 3 will be the optimal solution $\pi_{t+1}(.|s)$ to the original constrained optimization problem of the policy improvement step. Compute and write in simplified form its value function $V^{\pi_{t+1}}(s)$. Notice that in this setting, the state value function relates to the state-action value function through

$$V^{\pi}(s) = \mathbb{E}_{a\sim\mathcal{A}}[Q^{\pi}(s,a)] + \lambda\mathcal{H}(\pi(.|s)) \qquad \forall s \in \mathcal{S}.$$

Problem 3: In an imitation learning setting, we have collected demonstration data from an expert policy π_e, in particular, state-action pairs sampled from its normalized state-action occupancy measure, that is $(s,a) \sim \bar{\nu}_{\mu_0}^{\pi_e}$. The state space $\mathcal{S} = \{b, c, d, e\}$ and the action space $\mathcal{A} = \{x, y, z\}$. The data set consists of three data points:

$$\begin{aligned} \mathcal{D} = \{&(s^1 = b, a^1 = z) \\ &(s^2 = d, a^2 = y) \\ &(s^3 = b, a^3 = x).\} \end{aligned}$$

We would like to apply behavior cloning with the negative log-likelihood loss, that is, we aim to find $\hat{\pi}$ such that

$$\hat{\pi} = \arg\min_{\pi\in\Pi} \quad \mathbb{E}_{(s,a)\sim\bar{\nu}_{\mu_0}^{\pi_e}}\left[-\log\pi(a|s)\right],$$

where Π represents a class of softmax policies

$$\Pi = \{\pi_\theta : \pi_\theta(a|s) = \frac{\exp\left(\theta_{s,a}\right)}{\sum_{a'\in\mathcal{A}}\exp\left(\theta_{s,a'}\right)} \; for\ all\ s \in \mathcal{S}, a \in \mathcal{A}\ and\ \theta \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}\}.$$

Write the objective function $\mathcal{L}(\theta)$ of the empirical risk minimization to be used for fitting $\hat{\pi}$ to data set $\mathcal{D}$. The only variables in the objective function should be $\theta_{s,a}$ variables. [There is no need to solve the minimization problem.]

Problem 4: Consider the setting of reinforcement learning from human feedback. We would like to learn a reward function from preferences provided by a human over pairs of sub-trajectories. We follow the Bradley-Terry probabilistic model of human preference using discounted return, with discount factor $\gamma = 0.95$, as quality of a sub-trajectory. In the course of learning, we have arrived at a reward function $\hat{R}(s, a)$ with partial values listed below:

$$
\begin{aligned}
\hat{R}(empty, stay) &= -1, \quad \hat{R}(empty, explore) = 1, \\
\hat{R}(monster, stay) &= -2, \quad \hat{R}(monster, evade) = 3, \quad \hat{R}(monster, befriend) = -5, \\
\hat{R}(food, stay) &= -2, \quad \hat{R}(food, explore) = -2, \quad \hat{R}(food, collect) = 5, \\
\hat{R}(resource, stay) &= -2, \quad \hat{R}(resource, explore) = -2, \quad \hat{R}(resource, collect) = 4.
\end{aligned}
$$

Now, consider the following sub-trajectories:

$$
\begin{aligned}
&\tau_1 = (s_0 = empty, a_0 = stay, s_1 = monster, a_1 = evade, s_2 = resource, a_2 = collect) \\
&\tau_2 = (s_0 = empty, a_0 = explore, s_1 = food, a_1 = collect, s_2 = monster, a_2 = befriend) \\
&\tau_3 = (s_0 = empty, a_0 = explore, s_1 = empty, a_1 = explore, s_2 = resource, a_2 = collect)
\end{aligned}
$$

1. Compute the discounted return for each sub-trajectory τ_1, τ_2, and τ_3.
2. If we query the human with trajectory pair (τ_1, τ_2), with what probability will the human prefer τ_1 over τ_2 according to $\hat{R}$?
3. If we query the human with trajectory pair (τ_1, τ_3), with what probability will the human prefer τ_1 over τ_3 according to $\hat{R}$?
4. Suppose when queried about (τ_1, τ_2), the human picks τ_1 and when queried about (τ_1, τ_3), the human picks τ_3. Compute the (maximum likelihood estimation) loss function over these two data points according to $\hat{R}$.
5. If the human picks τ_2 over τ_1 and τ_3 over τ_1, how do you expect the loss function to change (decrease, stay the same, or increase) compared to Part 4? [Explain your reasoning without additional computation.]